Not Finished, But Abandoned

by

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ACKNOWLEDGMENTS

It is customary for authors of academic books to include in their prefaces statements such as this: "I am indebted to ... for their invaluable help; however, any errors which remain are my sole responsibility." Occasionally an author will go further. Rather than say that if there are any mistakes then he is responsible for them, he will say that there will inevitably be some mistakes and he is responsible for them....

Although the shouldering of all responsibility is usually a social ritual, the admission that errors exist is not — it is often a sincere avowal of belief. But this appears to present a living and everyday example of a situation which philosophers have commonly dismissed as absurd; that it is sometimes rational to hold logically incompatible beliefs.

— David C. Makinson (1965)

Above is the famous "preface paradox," which illustrates how to use the `wbepi` environment for epigraphs at the beginning of chapters. You probably also want to thank the Academy.

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LIST OF FIGURES

ABSTRACT

FIXME: basically a placeholder; do not believe

I did some research, read a bunch of papers, published a couple myself, (pick one):

1. ran some experiments and made some graphs,
2. proved some theorems

and now I have a job. I've assembled this document in the last couple of months so you will let me leave. Thanks!

1.1 What the heck is a plasma

1.1.1 Plasmas in the universe

1.1.2 How do plasmas affect us on earth?

1.1.3 Where does magnetic reconnection play into this whole shabang?

1.2 Magnetic Reconnection

1.2.1 The Generalized Ohm's Law

1.2.1.1 How do we derive this?

1.2.1.2 A breakdown of the terms

1.2.2 Regimes of Reconnection

1.2.2.1 Reconnection scaling parameters

i.e. why do we use S and L/d_i to characterize reconnection regimes?

Figure 1.1: Cartoon of places where reconnection can happen.

Figure 1.2: Diagram of reconnection regimes.

Figure 1.3: Diagram of Sweet-Parker reconnection.

Figure 1.4: Diagram of 2d hall reconnection

1.2.2.2 Where do various systems line up?

1.2.2.3 What are we interested in?

1.2.3 Sweet-Parker Reconnection

1.2.3.1 A quick historical detour

1.2.3.2 What are the assumptions that go into Sweet-Parker?

1.2.3.3 Sweet-Parker Reconnection Rate Derivation

1.2.3.4 Application to Real Systems

1.2.3.4.1 What are some example parameters where reconnection occurs?

1.2.3.4.2 What time scales does Sweet-Parker predict?

1.2.3.4.3 What did we miss?

1.2.4 Hall Reconnection

1.2.4.1 Why should the Hall term appear?

1.2.4.1.1 What is the relevant physics that sets when the hall term matters

1.2.4.1.2 In what kind of plasmas does the hall term appear and when does it not?

Figure 1.5: Flux tube changing size during reconnection causing pressure anisotropy build up.

Figure 1.6: Simulated kinetic reconnection with changing guide field.

1.2.5 Kinetic Reconnection

1.2.5.1 What makes kinetic reconnection "kinetic"?

1.2.5.1.1 What do collisions do?

1.2.5.1.1.1 How quickly do collisions damp out non-Maxwellian features?

1.2.5.1.2 What makes a system kinetic?

1.2.5.1.2.1 What parameters set this?

1.2.5.1.2.2 What should we do in the lab to get this?

1.2.5.2 Where does Pressure Anisotropy come from?

1.2.5.2.1 Movement of flux tube

1.2.5.2.2 Effect of background field

1.2.5.2.2.1 Too low of background field does...

1.2.5.2.2.2 Too high of field does...

1.2.5.2.2.3 Goldilocks!

Figure 1.7: Diagram from Ohia of with and without pressure anisotropy in simulation.

Figure 1.8: Some figure from Jan's papers thowing effects of pressure anisotropy

1.2.5.3 How does pressure anisotropy change what the reconnection looks like

1.2.5.3.1 Thinner reconnection layer? Use results from Jan's more recent papers.

1.2.5.3.2 Marginal Firehose Condition

1.2.5.3.3 Concluding: what features should we look for in the data?

1.2.5.4 Why do we want to measure it?

At the end of experimental setup I'll talk about why we want to do it in the lab but I think motivating why we care about kinetic reconnection as a process would be good here.

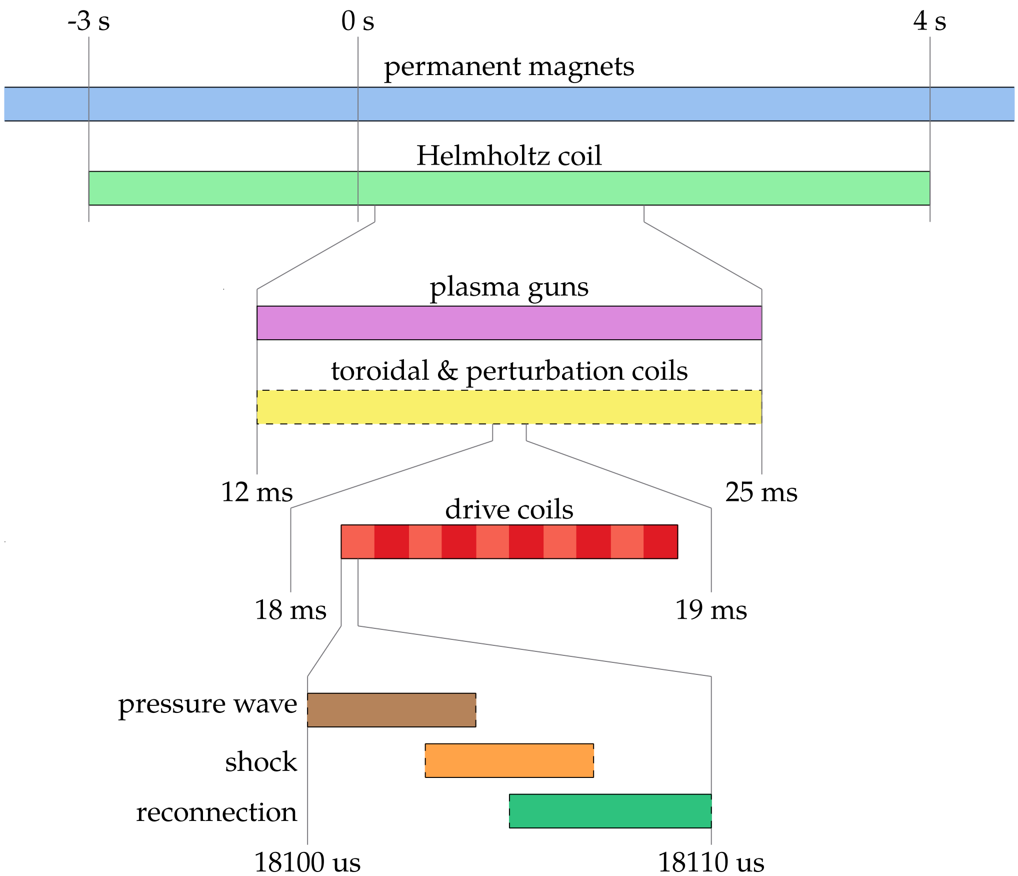

Figure 1.9: Timeline of experimental sequence.

Figure 1.10: Circuit diagram of plasma gun.

Figure 1.11: Simplified cut of plasma gun insides.

1.3 Experimental Setup

1.3.1 Overall how does our experiment progress

1.3.2 Initial field

1.3.2.1 Helmholtz

1.3.2.2 Permanent magnets

1.3.2.3 Toroidal

1.3.2.3.1 Shifted coil

1.3.2.3.2 Axial coil

1.3.2.4 Perturbation coils

1.3.3 Plasma Generation

1.3.3.1 How do our guns work

1.3.3.1.1 Gas lines

1.3.3.1.2 PFNs

1.3.3.1.3 Puff trigger

1.3.3.1.4 How to breakdown

Figure 1.12: Image from high speed camera of gun spewing

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1.3.3.2 How are the guns placed for injection?

1.3.3.2.1 How much do they fill the machine

1.3.3.2.2 What sits the filling fraction

1.3.3.3 What goes wrong with them

1.3.3.3.1 Burning up

1.3.3.3.2 Burn holes in wall because of background field Get image from Paul with hole in wall.

1.3.3.3.3 Air cooling

1.3.3.3.4 Optimizing gun operation

1.3.4 Drive field

1.3.4.1 Individual coils

1.3.4.1.1 Zone's between coils as plasma sources

1.3.4.2 Drive Cylinder

Reference Paul's RSI + thesis.

1.3.5 Diagnostics Overview

1.3.5.1 Bdot probes

1.3.5.1.1 Basic principles

1.3.5.1.2 Probe types

1.3.5.1.2.1 Linear probe

1.3.5.1.2.2 Hook probe

1.3.5.1.2.3 Speed probe

1.3.5.1.2.4 Flux array 1

1.3.5.1.2.5 Flux array 2

1.3.5.1.3 Data Interpretation How do we convert the raw data into something useful?

1.3.5.2 Hall probes

1.3.5.2.1 Basic principles

1.3.5.2.2 Issues and feasibility

1.3.5.3 Langmuir probes

1.3.5.3.1 Basic principles

1.3.5.3.2 What do we need to do to make them work for our device?

1.3.5.3.3 Look into XXX chapter

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1.3.5.4 Fast camera

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Figure 2.1: Greess et al. 2022 paper showing simulations with pressure wave.

3 MULTI-TIP LANGMUIR PROBE

4 PRESSURE ANISOTROPY PROBE

5 CONCLUSION

DISCARD THIS PAGE

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